Lab 7: Outliers

your name

Have students form groups of 2-3 people each. For each group, have them simulate data from a generative process like this.

- N is random (maybe poisson, centered at 500)
- P is fixed (maybe around 20)
- X is random and includes a constant (maybe a correlated standard normal, together with some categorical)
- β is drawn from a standard normal
- The residuals are homoskedastic and normal with random variance, so that the stdev of y is about twice that of βx

The groups will select a coefficient, β_k to estimate. Then have them modify or add (but not delete) 10 datapoints however they want to change $\hat{\beta}_k$ as much as possible, but in a way that is difficult to detect.

The groups then exchange *modified* datasets. The goal of the second group is to identify which points were added or modified and remove them. Having removed up to 10 points, they re-estimate $\hat{\beta}_k$. The goal is to get an estimate of $\hat{\beta}_k$ as close to the original as possible. Let's say that if the new estimate is within the standard 95% two-sided homoeskedastic CI, the second group wins.

Finally, have the groups come together. See whether they correctly identified the edited points, and how close they got to the truth. Ideally also share the results on the board, and have groups share their strategies.